

Practical 3 Naive Bayes Classifier

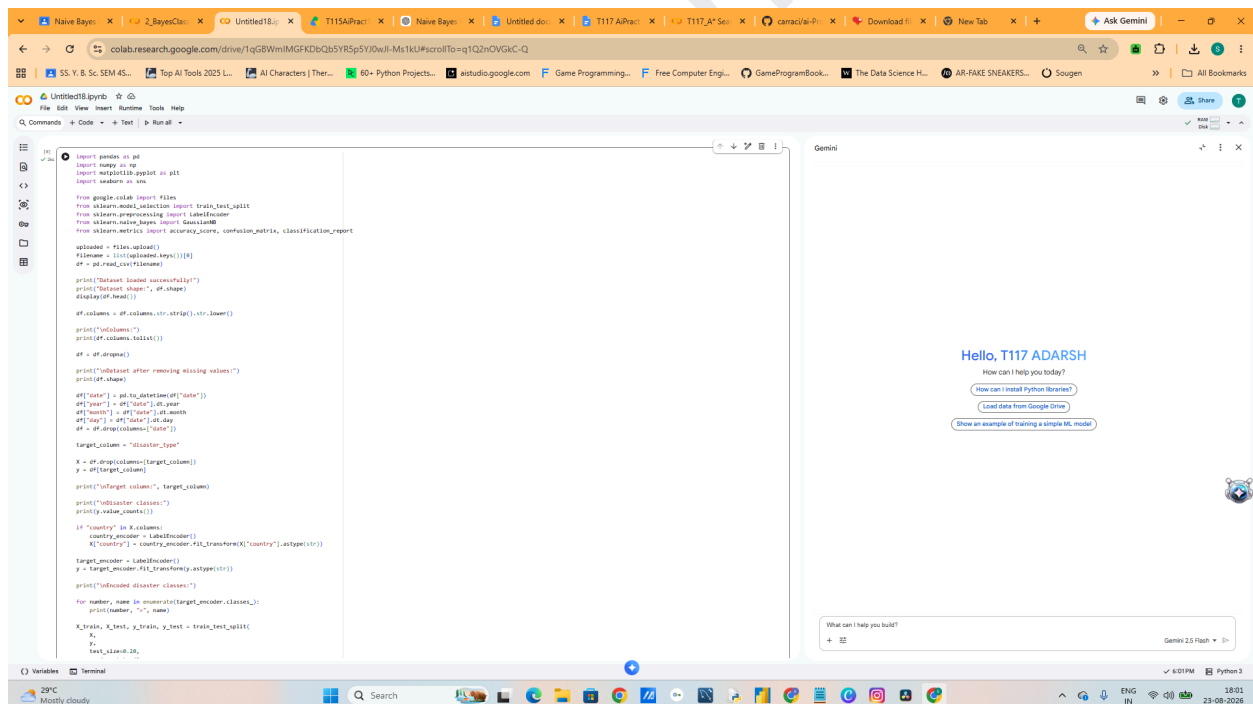
AIM : Naive Bayes Classifier

Implement the Naive Bayes' algorithm for classification.

Train a Naive Bayes' model using a given dataset and calculate class Probabilities.

Evaluate the accuracy of the model on test data and analyze the results.

Code :



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

uploaded = files.upload()
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)

print("Dataset loaded successfully")
print("Dataset shape:", df.shape)
display(df.head())

df.columns = df.columns.str.strip().str.lower()
print("\nColumns:")
print(df.columns.tolist())

df = df.dropna()

print("\nDataset after removing missing values")
print(df.shape)

df["date"] = pd.to_datetime(df["date"])
df["year"] = df["date"].dt.year
df["month"] = df["date"].dt.month
df["day"] = df["date"].dt.day
df = df.drop(columns=["date"])

target_column = "disaster_type"

X = df.drop(columns=[target_column])
y = df[target_column]

print("\nTarget column:", target_column)

print("\nDataset classes:")
print(y.value_counts())

if "country" in X.columns:
    country_encoder = LabelEncoder()
    X["country"] = country_encoder.fit_transform(X["country"].astype(str))

target_encoder = LabelEncoder()
y = target_encoder.fit_transform(y.astype(str))

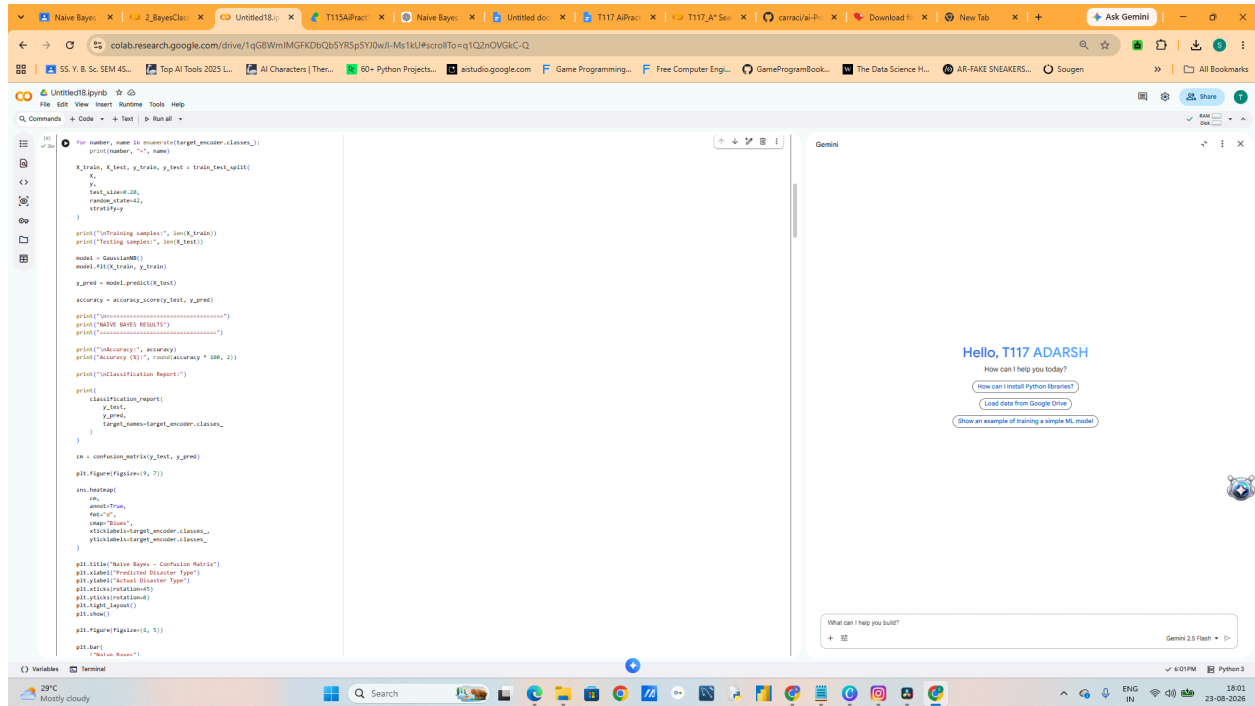
print("\nEncoded dataset classes:")

for number, name in enumerate(target_encoder.classes_):
    print(number, "-", name)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
```

Name : adarsh shukla
Roll no : T117

Practical 3 Naive Bayes Classifier



The screenshot shows a Jupyter Notebook environment with a code editor on the left and a Gemini chat interface on the right. The code in the notebook is as follows:

```
[1] ✓ 36 for number, name in enumerate(target_encoder.classes_):  
    print(number, "-", name)  
  
    X_train, X_test, y_train, y_test = train_test_split(  
        X,  
        y,  
        test_size=0.2,  
        random_state=42,  
        stratify=y  
    )  
  
    print("Training samples:", len(X_train))  
    print("Testing samples:", len(X_test))  
  
    model = GaussianNB()  
    model.fit(X_train, y_train)  
    y_pred = model.predict(X_test)  
  
    accuracy = accuracy_score(y_test, y_pred)  
  
    print("Accuracy: {}".format(accuracy))  
    print("NAIVE BAYES RESULTS")  
    print("Accuracy: {}".format(accuracy))  
  
    print("Labeling", accuracy)  
    print("Accuracy (3)", round(accuracy * 100, 2))  
    print("Classification Report")  
  
    print(classification_report(  
        y_test,  
        y_pred,  
        target_names=target_encoder.classes_,  
    ))  
  
    cm = confusion_matrix(y_test, y_pred)  
    plt.figure(figsize=(6, 7))  
    sns.heatmap(  
        cm,  
        annot=True,  
        fmt="r",  
        cbar=True,  
        xticklabels=target_encoder.classes_,  
        yticklabels=target_encoder.classes_,  
    )  
  
    plt.title("Naive Bayes - Confusion Matrix")  
    plt.xlabel("Predicted Classifier Type")  
    plt.ylabel("Actual Classifier Type")  
    plt.xticks(rotation=45)  
    plt.yticks(rotation=45)  
    plt.tight_layout()  
    plt.show()  
  
    plt.figure(figsize=(6, 5))  
  
    plt.bar(  
        ["Naive Bayes"],  
        [accuracy * 100],  
    )
```

Output :

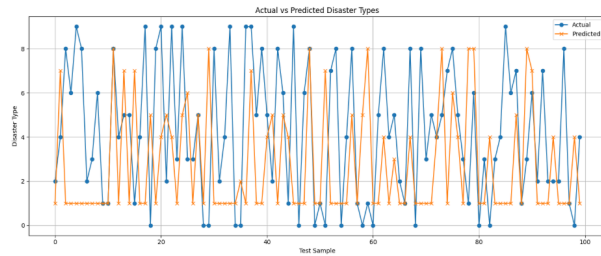
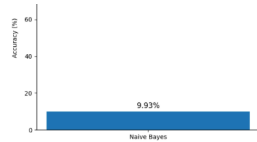
Name : adarsh shukla
Roll no : T117

Practical 3 Naive Bayes Classifier



Name : adarsh shukla
Roll no : T117

Practical 3 Naive Bayes Classifier



Actual vs Predicted:	Actual	Predicted
0	Extreme Heat	Earthquake
1	Hurricane	Tornado
2	Volcano Eruption	Earthquake
3	Storm Surge	Earthquake
4	Wildfire	Earthquake
5	Volcano Eruption	Earthquake
6	Extreme Heat	Earthquake
7	Flood	Earthquake
8	Storm Surge	Earthquake
9	Earthquake	Earthquake
10	Earthquake	Earthquake
11	Volcano Eruption	Volcano Eruption
12	Hurricane	Earthquake
13	Landslide	Tornado
14	Landslide	Earthquake
15	Earthquake	Tornado
16	Hurricane	Earthquake
17	Wildfire	Earthquake
18	Drought	Landslide
19	Volcano Eruption	Earthquake

